

GrounUp Enterprise Specification

Phase 05

Estimating & Cost Intelligence Engine

GES-P05-001 | Version 1.0.0 | REVIEW-READY

1. Executive Specification

Define the governed, explainable, configurable estimating and cost-intelligence platform for GrounUp SaaS, web, mobile, white-label, and OEM deployments.

1.1 Product Delivery Model

GrounUp shall be a multi-tenant SaaS platform. The primary office experience is a responsive web application. Offline-capable iOS and Android applications support field operations. Customer, vendor and agency portals provide limited role-specific access. White-label and OEM deployment are first-class capabilities rather than later add-ons.

1.2 User Experience and Customization

The interface shall remain simple through role-based workspaces, progressive disclosure, guided workflows, saved views and quick actions. Tenant administrators can configure terminology, services, cost codes, fields, forms, templates, approvals, dashboards, reports, notifications and AI behavior without modifying the shared core.

1.3 Governing Principles

- Explainable by default
- Configuration before customization
- Tenant isolation
- Role-based simplicity
- Human approval for material decisions
- Versioned calculations
- Offline-capable field capture
- API-first interoperability
- White-label readiness
- Auditable AI

2. Scope and Modules

Code	Module	Purpose
EST	Estimate Core	Estimate lifecycle, alternates, versions, approvals, lock/reopen, compare and export.
QTY	Quantity Intelligence	Manual, imported and AI-derived quantities with geometry, units, factors and confidence.
TAK	Takeoff	PDF/CAD/BIM/GIS/drone source ingestion, calibration, measurement, annotations and revision comparison.
PRD	Production	Production rates, crew/equipment balance, conditions, learning curves and historical calibration.
LAB	Labor Cost	Wages, burdens, overtime, union, travel, per diem and role productivity.
EQP	Equipment Cost	Owned/rented cost, utilization, fuel, maintenance, wear, attachments and standby.
MAT	Material Cost	Quotes, freight, waste, minimums, escalation, lead time and substitutions.
HAU	Hauling & Disposal	Trip cycles, payload, truck count, queues, tipping, swell/shrink and route constraints.
ASM	Assemblies	Reusable resource-driven work packages with formulas, options and constraints.
MUP	Markup & Margin	Overhead, contingency, bonds, tax, profit, fee, margin targets and bid rounding.
REG	Regional Pricing	Location factors, labor/equipment/material

		indices, vendor zones and effective dates.
RSK	Risk & Confidence	Risk register, uncertainty ranges, sensitivity, scenario and Monte Carlo specifications.
AI	AI Estimating Advisor	Document reading, recommendations, anomaly detection, explainability and approval gates.
CFG	Configuration Studio	No-code fields, forms, workflows, rules, terminology, themes and tenant overrides.
WLB	White Label & OEM	Branding, domains, sender identities, portal skinning, reseller hierarchy and app variants.
MOB	Field & Mobile	Offline capture, actual production, time, equipment, tickets, photos and sync.
API	Integration	Public APIs, webhooks, import/export and accounting/ERP/telematics links.
RPT	Reporting	Estimate review, variance, benchmark, profitability, audit and executive dashboards.

3. Functional Requirements

108 controlled requirements define the minimum review-ready scope. The complete register is included in CSV, JSON and Excel formats.

ID	Module	Title	Priority
GES-P05-REQ-001	EST	Estimate Core canonical model	Must
GES-P05-REQ-002	EST	Estimate Core configurable rules	Must
GES-P05-REQ-003	EST	Estimate Core auditability	Must
GES-P05-REQ-004	EST	Estimate Core permissions	Must
GES-P05-REQ-005	EST	Estimate Core import/export	Should
GES-P05-REQ-006	EST	Estimate Core validation and testing	Should
GES-P05-REQ-007	QTY	Quantity Intelligence canonical model	Must
GES-P05-REQ-008	QTY	Quantity Intelligence configurable rules	Must
GES-P05-REQ-009	QTY	Quantity Intelligence auditability	Must
GES-P05-REQ-010	QTY	Quantity Intelligence permissions	Must
GES-P05-REQ-011	QTY	Quantity Intelligence import/export	Should
GES-P05-REQ-012	QTY	Quantity Intelligence validation and testing	Should
GES-P05-REQ-013	TAK	Takeoff canonical model	Must
GES-P05-REQ-014	TAK	Takeoff configurable rules	Must
GES-P05-REQ-015	TAK	Takeoff auditability	Must
GES-P05-REQ-016	TAK	Takeoff permissions	Must
GES-P05-REQ-017	TAK	Takeoff import/export	Should
GES-P05-REQ-018	TAK	Takeoff validation and testing	Should
GES-P05-REQ-019	PRD	Production canonical model	Must
GES-P05-REQ-020	PRD	Production configurable rules	Must
GES-P05-REQ-021	PRD	Production auditability	Must
GES-P05-REQ-022	PRD	Production permissions	Must
GES-P05-REQ-023	PRD	Production import/export	Should
GES-P05-REQ-024	PRD	Production validation and testing	Should
GES-P05-REQ-025	LAB	Labor Cost canonical model	Must
GES-P05-REQ-026	LAB	Labor Cost configurable rules	Must
GES-P05-REQ-027	LAB	Labor Cost auditability	Must
GES-P05-REQ-028	LAB	Labor Cost permissions	Must
GES-P05-REQ-029	LAB	Labor Cost import/export	Should
GES-P05-REQ-030	LAB	Labor Cost validation and	Should

		testing	
GES-P05-REQ-031	EQP	Equipment Cost canonical model	Must
GES-P05-REQ-032	EQP	Equipment Cost configurable rules	Must
GES-P05-REQ-033	EQP	Equipment Cost auditability	Must
GES-P05-REQ-034	EQP	Equipment Cost permissions	Must
GES-P05-REQ-035	EQP	Equipment Cost import/export	Should
GES-P05-REQ-036	EQP	Equipment Cost validation and testing	Should
GES-P05-REQ-037	MAT	Material Cost canonical model	Must
GES-P05-REQ-038	MAT	Material Cost configurable rules	Must
GES-P05-REQ-039	MAT	Material Cost auditability	Must
GES-P05-REQ-040	MAT	Material Cost permissions	Must
GES-P05-REQ-041	MAT	Material Cost import/export	Should
GES-P05-REQ-042	MAT	Material Cost validation and testing	Should
GES-P05-REQ-043	HAU	Hauling & Disposal canonical model	Must
GES-P05-REQ-044	HAU	Hauling & Disposal configurable rules	Must
GES-P05-REQ-045	HAU	Hauling & Disposal auditability	Must

4. Canonical Data Model

Entity	Type	Definition
Estimate	Transactional	A governed pricing package for one project, opportunity or scenario.
EstimateVersion	Transactional	Immutable snapshot of an estimate at a point in time.
EstimateSection	Transactional	Hierarchical grouping of estimate content.
EstimateLine	Transactional	Priced work item with quantity, unit, cost and sell calculations.
EstimateAlternate	Transactional	Optional or comparative scope scenario.
EstimateAssumption	Transactional	Explicit assumption linked to affected calculations.
EstimateExclusion	Transactional	Explicitly excluded scope.
QuantityRecord	Transactional	Measured or calculated quantity with source lineage.
TakeoffDocument	Document	Plan/spec/source file used for takeoff.
TakeoffMeasurement	Transactional	Geometry and calculated quantity from a takeoff source.
TakeoffRevisionComparison	Analytical	Change comparison between document revisions.
Assembly	Master	Reusable work package composed of resources and formulas.
AssemblyComponent	Master	Labor, equipment, material, subcontract or other component.
ProductionRate	Master	Output per time with context and confidence.
ProductionConditionFactor	Master	Adjustment for weather, access, soil, phasing and other conditions.
CrewTemplate	Master	Reusable labor/equipment crew configuration.
LaborRate	Master	Base wage and burden components.
EquipmentRate	Master	Owned or rental rate and operating cost components.
MaterialPrice	Master	Material unit cost, vendor, freight and effective period.
VendorQuote	Document	Time-bound supplier or subcontractor quote.
TruckCycle	Analytical	Haul cycle inputs and calculated cycle time.
DisposalRate	Master	Facility fee and acceptance constraints.
FuelPrice	Master	Fuel price by type, location and effective

		period.
MarkupProfile	Master	Ordered markup and margin rules.
RegionalFactor	Master	Location adjustment by cost class and effective period.
RiskItem	Transactional	Estimate risk, probability, impact and mitigation.
Scenario	Analytical	Alternative set of inputs and outcomes.
SensitivityRun	Analytical	Impact of selected input changes.
MonteCarloRun	Analytical	Probabilistic simulation configuration and outcomes.
AIRecommendation	AI	Explainable AI proposal with evidence, confidence and disposition.
CalculationTrace	Audit	Input-to-output lineage for every computed value.
ApprovalStep	Workflow	Approval stage, assignee, decision and comments.
TenantEstimateConfiguration	Configuration	Tenant-specific settings, terminology and defaults.
WhiteLabelConfiguration	Configuration	Brand, domain, templates, sender identity and OEM settings.
CustomFieldDefinition	Configuration	No-code custom field definition.
CustomWorkflowDefinition	Configuration	No-code estimate workflow definition.
ActualProductionRecord	Transactional	Field-reported actual quantities, hours and resource usage.
CalibrationRecord	Analytical	Approved adjustment from historical actuals.
EstimateTemplate	Master	Reusable estimate structure and defaults.
CostCodeMapping	Master	Mapping to accounting, ERP, CSI, DOT or tenant codes.

5. Calculation Architecture

Quantity normalization

All measured and imported quantities are normalized through governed units while preserving source units.

Production calculation

Duration and resource hours are calculated from production rates, conditions, crew balance and constraints.

Cost build-up

Labor, equipment, material, subcontract, hauling, disposal and other costs remain separately traceable.

Sell-price calculation

Configured markups, margins, taxes, bonds, contingency and rounding are applied in explicit sequence.

Explainability

Every result retains source, formula, version, override, AI decision and approval lineage.

6. Business Rules

ID	Title	Rule
GES-P05-BR-001	Immutable approved versions	An approved estimate version shall not be edited; changes require a new version.
GES-P05-BR-002	Calculation trace required	Every sell amount shall retain traceable quantity, rate, formula, factors and markup sources.
GES-P05-BR-003	Tenant isolation	No tenant may access another tenant's

		estimates, pricing, actuals or configurations.
GES-P05-BR-004	AI requires disposition	Material AI recommendations shall be accepted, modified or rejected by an authorized user.
GES-P05-BR-005	Unit compatibility	Quantity and resource units shall be dimensionally compatible before calculation.
GES-P05-BR-006	Effective-date pricing	Pricing records shall be selected using estimate date and effective-date rules.
GES-P05-BR-007	Quote precedence	Approved project-specific quotes override regional and library prices for their scope.
GES-P05-BR-008	Markup sequence	Markup profiles shall define whether each component is additive, compounded or margin-targeted.
GES-P05-BR-009	No hidden overrides	Manual overrides shall require reason, author and timestamp.
GES-P05-BR-010	Crew balance	Crew output shall be constrained by the bottleneck resource unless an approved rule states otherwise.
GES-P05-BR-011	Haul cycle completeness	Truck calculations shall include load, travel, unload, return, queue and delay components.
GES-P05-BR-012	Swell and shrink separation	Bank, loose and compacted volumes shall not be treated as interchangeable.
GES-P05-BR-013	Confidence propagation	Estimate confidence shall consider source confidence, quantity confidence, rate confidence and risk.
GES-P05-BR-014	Historical calibration approval	Actual-based calibration shall not update master rates without review and approval.
GES-P05-BR-015	Role-based visibility	User workspaces shall expose only permitted modules, records and actions.
GES-P05-BR-016	White-label inheritance	OEM child tenants shall inherit allowed defaults while preserving approved local overrides.
GES-P05-BR-017	Offline conflict handling	Offline changes shall use deterministic conflict detection and review.
GES-P05-BR-018	Currency consistency	A single estimate version shall use one base currency with explicit conversion records when needed.
GES-P05-BR-019	Tax jurisdiction	Tax rules shall be selected from project jurisdiction and effective date.
GES-P05-BR-020	Estimate lock	Submitted estimates shall be locked from unapproved edits.
GES-P05-BR-021	Source retention	Referenced plans, quotes and attachments shall be retained according to policy.
GES-P05-BR-022	Scenario isolation	Scenario changes shall not alter the baseline estimate until promoted.
GES-P05-BR-023	Rounding policy	Rounding shall occur only at configured stages and retain unrounded calculation values.
GES-P05-BR-024	Negative cost control	Negative quantities or rates require an approved adjustment type.
GES-P05-BR-025	Deletion policy	Used master records shall be retired, not deleted.
GES-P05-BR-026	Regional override priority	Tenant and project overrides shall follow an explicit precedence hierarchy.
GES-P05-BR-027	Production factor bounds	Condition factors outside configured bounds require approval.
GES-P05-BR-028	Fuel model selection	Equipment fuel may use measured, manufacturer, historical or formula sources with source type recorded.
GES-P05-BR-029	Bid audit export	The system shall produce a review package containing assumptions, exceptions and calculation lineage.
GES-P05-BR-030	Template governance	Published templates require version, owner and approval status.

7. Engines

ID	Engine	Purpose
GES-P05-ENG-001	Quantity Engine	Converts measured geometry and formulas into normalized quantities.
GES-P05-ENG-002	Takeoff Intelligence Engine	Reads and compares plans/specifications with human review.
GES-P05-ENG-003	Production Engine	Calculates duration and hours using rates, crews and conditions.
GES-P05-ENG-004	Crew Balancer	Identifies bottlenecks and recommends balanced resource mixes.
GES-P05-ENG-005	Labor Cost Engine	Builds fully burdened labor cost.
GES-P05-ENG-006	Equipment Cost Engine	Builds ownership/rental and operating costs.
GES-P05-ENG-007	Material Cost Engine	Selects and normalizes material pricing.
GES-P05-ENG-008	Haul & Disposal Engine	Calculates truck cycles, counts, haul cost and disposal.
GES-P05-ENG-009	Markup & Margin Engine	Applies overhead, contingency, tax, bonds, profit and margin rules.
GES-P05-ENG-010	Regional Pricing Engine	Applies location and market factors.
GES-P05-ENG-011	Risk & Confidence Engine	Scores uncertainty and confidence.
GES-P05-ENG-012	Scenario Engine	Maintains isolated alternatives and comparisons.
GES-P05-ENG-013	Simulation Engine	Runs probabilistic cost and duration simulations.
GES-P05-ENG-014	Explainability Engine	Produces calculation lineage and plain-language explanations.
GES-P05-ENG-015	Calibration Engine	Compares estimates with actuals and proposes governed updates.
GES-P05-ENG-016	Estimate Validation Engine	Runs completeness, consistency and policy checks.
GES-P05-ENG-017	Document Classification Engine	Classifies plans, specs, addenda, reports and quotes.
GES-P05-ENG-018	White-Label Configuration Engine	Resolves inherited brand and tenant configuration.
GES-P05-ENG-019	Offline Sync Engine	Synchronizes field records with conflict handling.
GES-P05-ENG-020	Export Composition Engine	Builds proposals, review books and machine-readable exports.

8. AI Estimating Advisor

The AI assistant may read plans, specifications, addenda, reports, quotes and historical records; create reviewable draft quantities and line items; recommend crew size, equipment mix, fuel, trucks, duration, assumptions, questions and risks; and explain cost drivers. It may not silently approve, submit, alter financial terms, overwrite governed libraries or publish calibration changes.

9. White Label, OEM and No-Code Configuration

Branding includes company name, logo, colors, domain, login screen, email sender, proposal/report templates, customer portal and mobile application identity. OEM deployments support parent-child tenant inheritance, reseller administration, plan packaging and controlled local overrides. Configuration Studio supports fields, forms, workflows, terminology, validations, dashboards, reports, templates and AI policies.

10. Workflows

ID	Workflow	Description
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GES-P05-WF-001	Estimate Creation	Draft estimate from template, opportunity or blank.
GES-P05-WF-002	Document-to-Estimate	Upload documents, classify, extract, review quantities and create draft lines.
GES-P05-WF-003	Takeoff Revision Review	Compare plan revisions and disposition detected changes.
GES-P05-WF-004	Vendor Quote Intake	Capture quote, map items, validate dates and approve use.
GES-P05-WF-005	Estimate Review and Approval	Estimator review, operations review, executive approval and submission.
GES-P05-WF-006	AI Recommendation Disposition	Review AI evidence, accept/modify/reject and log rationale.
GES-P05-WF-007	Scenario Comparison	Create alternatives, run calculations and promote selected scenario.
GES-P05-WF-008	Historical Calibration	Compare estimated versus actual production and approve updates.
GES-P05-WF-009	White-Label Provisioning	Configure brand, domain, templates, sender identity and tenant inheritance.
GES-P05-WF-010	Offline Field Sync	Capture actuals offline, sync, resolve conflicts and post approved records.
GES-P05-WF-011	Estimate Reopen	Request reopening, approve, clone version and preserve submitted version.
GES-P05-WF-012	Change Order Pricing	Create estimate from approved project change and route approval.
GES-P05-WF-013	Bid Export	Generate proposal, internal review book, schedule and audit package.
GES-P05-WF-014	Rate Expiration Review	Identify expiring rates and route owners for renewal.
GES-P05-WF-015	Risk Simulation	Define uncertainty, run scenario/simulation and review results.

11. API Surface

Method	Path	Purpose
POST	/v1/estimates	Create estimate
GET	/v1/estimates/{id}	Read estimate
POST	/v1/estimates/{id}/versions	Create immutable version
POST	/v1/estimates/{id}/calculate	Calculate estimate
POST	/v1/estimates/{id}/validate	Validate estimate
POST	/v1/estimates/{id}/submit	Submit for approval
POST	/v1/estimates/{id}/approve	Approve estimate
POST	/v1/estimates/{id}/scenarios	Create scenario
POST	/v1/takeoff/documents	Upload takeoff document
POST	/v1/takeoff/measurements	Create measurement
POST	/v1/takeoff/revision-comparisons	Compare revisions
GET	/v1/assemblies	List assemblies
POST	/v1/production/calculate	Calculate production
POST	/v1/haul/calculate	Calculate hauling
POST	/v1/risk/simulations	Run risk simulation
GET	/v1/calculations/{trace_id}	Read calculation trace
POST	/v1/ai/recommendations/{id}/disposition	Disposition AI recommendation
GET	/v1/configuration/tenant	Read tenant configuration
PUT	/v1/configuration/white-label	Update white-label configuration
POST	/v1/actuals	Post actual production
POST	/v1/calibration/proposals	Create calibration proposal
GET	/v1/reports/estimate-review/{id}	Generate review report
POST	/v1/exports/estimate/{id}	Create export package
POST	/v1/webhooks	Register webhook

12. UI/UX Surfaces

Surface	Channel	Role	Description
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Estimate Workspace	Web	Estimator	Tree, grid, detail panel, calculation trace and AI assistant.
Quick Estimate	Web/Mobile	Owner	Guided wizard for small contractors.
Takeoff Studio	Web	Estimator	Document viewer, calibration, measurements, layers and revision compare.
Cost Library Browser	Web	Estimator/Admin	Search, filter, compare and approve rates.
Crew & Production Designer	Web	Estimator/Operations	Visual crew balance, bottleneck and duration modeling.
Scenario Compare	Web	Estimator/Executive	Side-by-side cost, duration, risk and margin comparison.
Estimate Review Center	Web/Tablet	Reviewer	Exceptions, assumptions, overrides, AI decisions and approvals.
Field Actuals	Mobile	Foreman	Offline time, quantity, equipment, tickets, photos and notes.
Configuration Studio	Web	Tenant Admin	No-code fields, forms, workflows, rules, terminology and defaults.
Brand Studio	Web	Tenant/OEM Admin	Logo, theme, domain, emails, reports, portals and mobile branding.
Executive Estimating Dashboard	Web	Executive	Pipeline, win rate, margin, variance, risk and estimator performance.
Customer Proposal Portal	Responsive Web	Customer	Review scope, alternates, documents and electronic acceptance.

13. Reporting

Report	Purpose
Estimate Summary	Scope, direct cost, sell, margin and alternates.
Detailed Cost Build-Up	Resource-level cost and markup detail.
Estimate Review Book	Assumptions, exclusions, risks, overrides and approvals.
Calculation Trace Report	Every amount with source and formula lineage.
Production & Duration Report	Crews, equipment, outputs, hours and schedule assumptions.
Haul Analysis	Cycle time, truck count, payload, queues and cost.
Risk & Confidence Report	Uncertainty, sensitivity and confidence drivers.
Scenario Comparison	Baseline versus alternatives.
Vendor Quote Comparison	Normalized quote comparison and exceptions.
Historical Estimate vs Actual	Cost, quantity, hours and production variance.
Calibration Proposal Report	Suggested rate updates with evidence.
Regional Price Variance	Local versus baseline prices and factors.
Bid Margin Waterfall	Direct cost through final sell price.
White-Label Configuration Audit	Inherited and overridden branding/configuration.
Executive Estimating KPI	Pipeline, win rate, cycle time, margin and accuracy.

14. Security and Nonfunctional Requirements

- Strict tenant isolation and least-privilege access
- Encryption in transit and at rest
- Audit logging for reads of sensitive financial data and all writes
- Configurable retention, legal hold and archive
- Performance targets for estimate grids and calculation services
- Offline sync with deterministic conflict resolution

- Accessible web and mobile experiences
- Disaster recovery, backups and regional deployment options
- API rate limits, idempotency and webhook signing
- Model, prompt and recommendation version logging

15. Testing and Acceptance

Each requirement has a mapped test case. Release acceptance requires calculation regression, unit compatibility, tenant-isolation, security, usability, accessibility, offline sync, export fidelity, load/performance and AI-evaluation evidence.

16. Phase Exit Criteria

- All artifacts reviewed
- Critical issues resolved
- Requirements traceability complete
- Architecture and security approved
- Estimating SME approval
- AI governance approval
- Version 1.0.0 tagged and locked

Appendix A - Requirement Register (continued)

ID	Module	Title	Acceptance
GES-P05-REQ-046	HAU	Hauling & Disposal permissions	Demonstrate hauling & disposal permissions with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-047	HAU	Hauling & Disposal import/export	Demonstrate hauling & disposal import/export with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-048	HAU	Hauling & Disposal validation and testing	Demonstrate hauling & disposal validation and testing with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-049	ASM	Assemblies canonical model	Demonstrate assemblies canonical model with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-050	ASM	Assemblies configurable rules	Demonstrate assemblies configurable rules with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-051	ASM	Assemblies auditability	Demonstrate assemblies auditability with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-052	ASM	Assemblies permissions	Demonstrate assemblies permissions with tenant-specific

			configuration, role controls, version history, and traceable output.
GES-P05-REQ-053	ASM	Assemblies import/export	Demonstrate assemblies import/export with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-054	ASM	Assemblies validation and testing	Demonstrate assemblies validation and testing with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-055	MUP	Markup & Margin canonical model	Demonstrate markup & margin canonical model with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-056	MUP	Markup & Margin configurable rules	Demonstrate markup & margin configurable rules with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-057	MUP	Markup & Margin auditability	Demonstrate markup & margin auditability with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-058	MUP	Markup & Margin permissions	Demonstrate markup & margin permissions with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-059	MUP	Markup & Margin import/export	Demonstrate markup & margin import/export with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-060	MUP	Markup & Margin validation and testing	Demonstrate markup & margin validation and testing with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-061	REG	Regional Pricing canonical model	Demonstrate regional pricing canonical model with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-062	REG	Regional Pricing configurable rules	Demonstrate regional pricing configurable rules with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-063	REG	Regional Pricing auditability	Demonstrate regional pricing auditability with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-064	REG	Regional Pricing permissions	Demonstrate regional pricing permissions with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-065	REG	Regional Pricing import/export	Demonstrate regional pricing import/export with tenant-specific configuration, role controls, version history, and

			traceable output.
GES-P05-REQ-066	REG	Regional Pricing validation and testing	Demonstrate regional pricing validation and testing with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-067	RSK	Risk & Confidence canonical model	Demonstrate risk & confidence canonical model with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-068	RSK	Risk & Confidence configurable rules	Demonstrate risk & confidence configurable rules with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-069	RSK	Risk & Confidence auditability	Demonstrate risk & confidence auditability with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-070	RSK	Risk & Confidence permissions	Demonstrate risk & confidence permissions with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-071	RSK	Risk & Confidence import/export	Demonstrate risk & confidence import/export with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-072	RSK	Risk & Confidence validation and testing	Demonstrate risk & confidence validation and testing with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-073	AI	AI Estimating Advisor canonical model	Demonstrate ai estimating advisor canonical model with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-074	AI	AI Estimating Advisor configurable rules	Demonstrate ai estimating advisor configurable rules with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-075	AI	AI Estimating Advisor auditability	Demonstrate ai estimating advisor auditability with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-076	AI	AI Estimating Advisor permissions	Demonstrate ai estimating advisor permissions with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-077	AI	AI Estimating Advisor import/export	Demonstrate ai estimating advisor import/export with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-078	AI	AI Estimating Advisor validation and testing	Demonstrate ai estimating advisor validation and testing with tenant-specific configuration, role controls, version history, and traceable output.

GES-P05-REQ-079	CFG	Configuration Studio canonical model	Demonstrate configuration studio canonical model with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-080	CFG	Configuration Studio configurable rules	Demonstrate configuration studio configurable rules with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-081	CFG	Configuration Studio auditability	Demonstrate configuration studio auditability with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-082	CFG	Configuration Studio permissions	Demonstrate configuration studio permissions with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-083	CFG	Configuration Studio import/export	Demonstrate configuration studio import/export with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-084	CFG	Configuration Studio validation and testing	Demonstrate configuration studio validation and testing with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-085	WLB	White Label & OEM canonical model	Demonstrate white label & oem canonical model with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-086	WLB	White Label & OEM configurable rules	Demonstrate white label & oem configurable rules with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-087	WLB	White Label & OEM auditability	Demonstrate white label & oem auditability with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-088	WLB	White Label & OEM permissions	Demonstrate white label & oem permissions with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-089	WLB	White Label & OEM import/export	Demonstrate white label & oem import/export with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-090	WLB	White Label & OEM validation and testing	Demonstrate white label & oem validation and testing with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-091	MOB	Field & Mobile canonical model	Demonstrate field & mobile canonical model with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-092	MOB	Field & Mobile configurable	Demonstrate field & mobile

		rules	configurable rules with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-093	MOB	Field & Mobile auditability	Demonstrate field & mobile auditability with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-094	MOB	Field & Mobile permissions	Demonstrate field & mobile permissions with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-095	MOB	Field & Mobile import/export	Demonstrate field & mobile import/export with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-096	MOB	Field & Mobile validation and testing	Demonstrate field & mobile validation and testing with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-097	API	Integration canonical model	Demonstrate integration canonical model with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-098	API	Integration configurable rules	Demonstrate integration configurable rules with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-099	API	Integration auditability	Demonstrate integration auditability with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-100	API	Integration permissions	Demonstrate integration permissions with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-101	API	Integration import/export	Demonstrate integration import/export with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-102	API	Integration validation and testing	Demonstrate integration validation and testing with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-103	RPT	Reporting canonical model	Demonstrate reporting canonical model with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-104	RPT	Reporting configurable rules	Demonstrate reporting configurable rules with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-105	RPT	Reporting auditability	Demonstrate reporting auditability with tenant-specific configuration, role controls,

			version history, and traceable output.
GES-P05-REQ-106	RPT	Reporting permissions	Demonstrate reporting permissions with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-107	RPT	Reporting import/export	Demonstrate reporting import/export with tenant-specific configuration, role controls, version history, and traceable output.
GES-P05-REQ-108	RPT	Reporting validation and testing	Demonstrate reporting validation and testing with tenant-specific configuration, role controls, version history, and traceable output.

Appendix B - Validation Rules

ID	Field	Rule	Message
GES-P05-VR-001	Estimate.name	Required and 3-200 characters	Enter an estimate name.
GES-P05-VR-002	Estimate.currency	ISO 4217 code	Select a valid currency.
GES-P05-VR-003	EstimateLine.quantity	Numeric and not null	Enter a valid quantity.
GES-P05-VR-004	EstimateLine.unit	Active unit and dimension-compatible	Select a compatible unit.
GES-P05-VR-005	EstimateLine.unit_cost	Numeric; negative requires adjustment type	Enter a valid unit cost.
GES-P05-VR-006	ProductionRate.output	Greater than zero	Production output must be greater than zero.
GES-P05-VR-007	LaborRate.effective_from	Required date	Enter an effective date.
GES-P05-VR-008	EquipmentRate.rate_basis	Hourly, daily, weekly, monthly or unit	Select a valid rate basis.
GES-P05-VR-009	MaterialPrice.vendor_id	Required when source type is vendor	Select the vendor.
GES-P05-VR-010	TruckCycle.payload	Greater than zero and within vehicle limit	Enter a valid payload.
GES-P05-VR-011	TruckCycle.distance	Nonnegative	Distance cannot be negative.
GES-P05-VR-012	MarkupProfile.sequence	Unique positive integers	Correct markup sequence.
GES-P05-VR-013	RegionalFactor.factor	Within configured min/max	Regional factor is outside allowed range.
GES-P05-VR-014	AIRecommendation.confidence	0-100	Confidence must be 0 through 100.
GES-P05-VR-015	ApprovalStep.decision	Approved, Rejected, Returned or Pending	Select a valid decision.
GES-P05-VR-016	WhiteLabelConfiguration.domain	Unique verified domain when enabled	Verify and assign a unique domain.
GES-P05-VR-017	CustomFieldDefinition.data_type	Supported field type	Select a supported data type.
GES-P05-VR-018	ActualProductionRecord.work_date	Not future beyond policy	Enter a valid work date.
GES-P05-VR-019	EstimateVersion.version_no	Unique within estimate	Version number already exists.
GES-P05-VR-020	VendorQuote.expiration_date	On or after issue date	Quote expiration must follow issue date.

Appendix C - Traceability Summary

Requirement	Entity	Rule	Engine	API	UI	Test
GES-P05-REQ-001	Estimate	GES-P05-BR-001	GES-P05-ENG-016	GES-P05-API-001	GES-P05-UI-001	GES-P05-TC-001
GES-P05-REQ-002	EstimateVersion	GES-P05-BR-002	GES-P05-ENG-016	GES-P05-API-002	GES-P05-UI-002	GES-P05-TC-002
GES-P05-REQ-003	EstimateSection	GES-P05-BR-003	GES-P05-	GES-P05-API-003	GES-P05-UI-003	GES-P05-TC-003

			ENG-016			
GES-P05-REQ-004	EstimateLine	GES-P05-BR-004	GES-P05-ENG-016	GES-P05-API-004	GES-P05-UI-004	GES-P05-TC-004
GES-P05-REQ-005	EstimateAlternate	GES-P05-BR-005	GES-P05-ENG-016	GES-P05-API-005	GES-P05-UI-005	GES-P05-TC-005
GES-P05-REQ-006	EstimateAssumption	GES-P05-BR-006	GES-P05-ENG-016	GES-P05-API-006	GES-P05-UI-006	GES-P05-TC-006
GES-P05-REQ-007	EstimateExclusion	GES-P05-BR-007	GES-P05-ENG-007	GES-P05-API-007	GES-P05-UI-007	GES-P05-TC-007
GES-P05-REQ-008	QuantityRecord	GES-P05-BR-008	GES-P05-ENG-008	GES-P05-API-008	GES-P05-UI-008	GES-P05-TC-008
GES-P05-REQ-009	TakeoffDocument	GES-P05-BR-009	GES-P05-ENG-009	GES-P05-API-009	GES-P05-UI-009	GES-P05-TC-009
GES-P05-REQ-010	TakeoffMeasurement	GES-P05-BR-010	GES-P05-ENG-010	GES-P05-API-010	GES-P05-UI-010	GES-P05-TC-010
GES-P05-REQ-011	TakeoffRevisionComparison	GES-P05-BR-011	GES-P05-ENG-011	GES-P05-API-011	GES-P05-UI-011	GES-P05-TC-011
GES-P05-REQ-012	Assembly	GES-P05-BR-012	GES-P05-ENG-012	GES-P05-API-012	GES-P05-UI-012	GES-P05-TC-012
GES-P05-REQ-013	AssemblyComponent	GES-P05-BR-013	GES-P05-ENG-002	GES-P05-API-013	GES-P05-UI-001	GES-P05-TC-013
GES-P05-REQ-014	ProductionRate	GES-P05-BR-014	GES-P05-ENG-002	GES-P05-API-014	GES-P05-UI-002	GES-P05-TC-014
GES-P05-REQ-015	ProductionConditionFactor	GES-P05-BR-015	GES-P05-ENG-002	GES-P05-API-015	GES-P05-UI-003	GES-P05-TC-015
GES-P05-REQ-016	CrewTemplate	GES-P05-BR-016	GES-P05-ENG-002	GES-P05-API-016	GES-P05-UI-004	GES-P05-TC-016
GES-P05-REQ-017	LaborRate	GES-P05-BR-017	GES-P05-ENG-002	GES-P05-API-017	GES-P05-UI-005	GES-P05-TC-017
GES-P05-REQ-018	EquipmentRate	GES-P05-BR-018	GES-P05-ENG-002	GES-P05-API-018	GES-P05-UI-006	GES-P05-TC-018
GES-P05-REQ-019	MaterialPrice	GES-P05-BR-019	GES-P05-ENG-019	GES-P05-API-019	GES-P05-UI-007	GES-P05-TC-019
GES-P05-REQ-020	VendorQuote	GES-P05-BR-020	GES-P05-ENG-020	GES-P05-API-020	GES-P05-UI-008	GES-P05-TC-020
GES-P05-REQ-021	TruckCycle	GES-P05-BR-021	GES-P05-ENG-001	GES-P05-API-021	GES-P05-UI-009	GES-P05-TC-021
GES-P05-REQ-022	DisposalRate	GES-P05-BR-022	GES-P05-ENG-002	GES-P05-API-022	GES-P05-UI-010	GES-P05-TC-022
GES-P05-REQ-023	FuelPrice	GES-P05-BR-023	GES-P05-ENG-003	GES-P05-API-023	GES-P05-UI-011	GES-P05-TC-023
GES-P05-REQ-024	MarkupProfile	GES-P05-BR-024	GES-P05-ENG-004	GES-P05-API-024	GES-P05-UI-012	GES-P05-TC-024
GES-P05-REQ-025	RegionalFactor	GES-P05-BR-025	GES-P05-ENG-005	GES-P05-API-001	GES-P05-UI-001	GES-P05-TC-025
GES-P05-REQ-026	RiskItem	GES-P05-BR-026	GES-P05-ENG-005	GES-P05-API-002	GES-P05-UI-002	GES-P05-TC-026
GES-P05-REQ-027	Scenario	GES-P05-BR-027	GES-P05-ENG-005	GES-P05-API-003	GES-P05-UI-003	GES-P05-TC-027
GES-P05-REQ-028	SensitivityRun	GES-P05-BR-028	GES-P05-ENG-005	GES-P05-API-004	GES-P05-UI-004	GES-P05-TC-028
GES-P05-REQ-029	MonteCarloRun	GES-P05-BR-029	GES-P05-ENG-005	GES-P05-API-005	GES-P05-UI-005	GES-P05-TC-029
GES-P05-REQ-030	AIRecommendation	GES-P05-BR-030	GES-P05-ENG-005	GES-P05-API-006	GES-P05-UI-006	GES-P05-TC-030
GES-P05-REQ-031	CalculationTrace	GES-P05-BR-001	GES-P05-ENG-011	GES-P05-API-007	GES-P05-UI-007	GES-P05-TC-031
GES-P05-REQ-032	ApprovalStep	GES-P05-BR-002	GES-P05-ENG-012	GES-P05-API-008	GES-P05-UI-008	GES-P05-TC-032
GES-P05-REQ-033	TenantEstimateConfiguration	GES-P05-BR-003	GES-P05-ENG-013	GES-P05-API-009	GES-P05-UI-009	GES-P05-TC-033
GES-P05-REQ-034	WhiteLabelConfiguration	GES-P05-BR-004	GES-P05-ENG-014	GES-P05-API-010	GES-P05-UI-010	GES-P05-TC-034
GES-P05-REQ-035	CustomFieldDefinition	GES-P05-BR-005	GES-P05-ENG-015	GES-P05-API-011	GES-P05-UI-011	GES-P05-TC-035
GES-P05-REQ-036	CustomWorkflo	GES-P05-BR-006	GES-P05-	GES-P05-API-012	GES-P05-UI-012	GES-P05-TC-036

	wDefinition		ENG-016			
GES-P05-REQ-037	ActualProductionRecord	GES-P05-BR-007	GES-P05-ENG-007	GES-P05-API-013	GES-P05-UI-001	GES-P05-TC-037
GES-P05-REQ-038	CalibrationRecord	GES-P05-BR-008	GES-P05-ENG-007	GES-P05-API-014	GES-P05-UI-002	GES-P05-TC-038
GES-P05-REQ-039	EstimateTemplate	GES-P05-BR-009	GES-P05-ENG-007	GES-P05-API-015	GES-P05-UI-003	GES-P05-TC-039
GES-P05-REQ-040	CostCodeMapping	GES-P05-BR-010	GES-P05-ENG-007	GES-P05-API-016	GES-P05-UI-004	GES-P05-TC-040
GES-P05-REQ-041	Estimate	GES-P05-BR-011	GES-P05-ENG-007	GES-P05-API-017	GES-P05-UI-005	GES-P05-TC-041
GES-P05-REQ-042	EstimateVersion	GES-P05-BR-012	GES-P05-ENG-007	GES-P05-API-018	GES-P05-UI-006	GES-P05-TC-042
GES-P05-REQ-043	EstimateSection	GES-P05-BR-013	GES-P05-ENG-008	GES-P05-API-019	GES-P05-UI-007	GES-P05-TC-043
GES-P05-REQ-044	EstimateLine	GES-P05-BR-014	GES-P05-ENG-008	GES-P05-API-020	GES-P05-UI-008	GES-P05-TC-044
GES-P05-REQ-045	EstimateAlternate	GES-P05-BR-015	GES-P05-ENG-008	GES-P05-API-021	GES-P05-UI-009	GES-P05-TC-045
GES-P05-REQ-046	EstimateAssumption	GES-P05-BR-016	GES-P05-ENG-008	GES-P05-API-022	GES-P05-UI-010	GES-P05-TC-046
GES-P05-REQ-047	EstimateExclusion	GES-P05-BR-017	GES-P05-ENG-008	GES-P05-API-023	GES-P05-UI-011	GES-P05-TC-047
GES-P05-REQ-048	QuantityRecord	GES-P05-BR-018	GES-P05-ENG-008	GES-P05-API-024	GES-P05-UI-012	GES-P05-TC-048
GES-P05-REQ-049	TakeoffDocument	GES-P05-BR-019	GES-P05-ENG-009	GES-P05-API-001	GES-P05-UI-001	GES-P05-TC-049
GES-P05-REQ-050	TakeoffMeasurement	GES-P05-BR-020	GES-P05-ENG-010	GES-P05-API-002	GES-P05-UI-002	GES-P05-TC-050
GES-P05-REQ-051	TakeoffRevisionComparison	GES-P05-BR-021	GES-P05-ENG-011	GES-P05-API-003	GES-P05-UI-003	GES-P05-TC-051
GES-P05-REQ-052	Assembly	GES-P05-BR-022	GES-P05-ENG-012	GES-P05-API-004	GES-P05-UI-004	GES-P05-TC-052
GES-P05-REQ-053	AssemblyComponent	GES-P05-BR-023	GES-P05-ENG-013	GES-P05-API-005	GES-P05-UI-005	GES-P05-TC-053
GES-P05-REQ-054	ProductionRate	GES-P05-BR-024	GES-P05-ENG-014	GES-P05-API-006	GES-P05-UI-006	GES-P05-TC-054
GES-P05-REQ-055	ProductionConditionFactor	GES-P05-BR-025	GES-P05-ENG-015	GES-P05-API-007	GES-P05-UI-007	GES-P05-TC-055
GES-P05-REQ-056	CrewTemplate	GES-P05-BR-026	GES-P05-ENG-016	GES-P05-API-008	GES-P05-UI-008	GES-P05-TC-056
GES-P05-REQ-057	LaborRate	GES-P05-BR-027	GES-P05-ENG-017	GES-P05-API-009	GES-P05-UI-009	GES-P05-TC-057
GES-P05-REQ-058	EquipmentRate	GES-P05-BR-028	GES-P05-ENG-018	GES-P05-API-010	GES-P05-UI-010	GES-P05-TC-058
GES-P05-REQ-059	MaterialPrice	GES-P05-BR-029	GES-P05-ENG-019	GES-P05-API-011	GES-P05-UI-011	GES-P05-TC-059
GES-P05-REQ-060	VendorQuote	GES-P05-BR-030	GES-P05-ENG-020	GES-P05-API-012	GES-P05-UI-012	GES-P05-TC-060